The Decryption Key: How ShashChess is Unlocking the Black Box of Neural Network Chess Engines

Section 1: The New Oracle of Chess: Power, Mystery, and the Black Box Problem

The world of chess has been irrevocably transformed by the advent of artificial intelligence. For decades, computer engines served as powerful sparring partners and analytical tools, their strength growing in a steady, comprehensible manner. This evolution, however, has recently undergone a revolution. The introduction of neural network-based evaluation functions, specifically the NNUE (Efficiently Updatable Neural Network) architecture pioneered in Shogi and later adopted by engines like Stockfish, has created a new class of chess-playing entity.[1] These engines are not merely stronger than their predecessors; they represent a fundamental paradigm shift in how a machine perceives and plays the game. Their dominance is absolute, consistently conquering the most prestigious engine tournaments and establishing a level of play that is, without exaggeration, otherworldly.[1]

This leap in performance, however, has come at a cost: transparency. While classical engines were complex, their logic was ultimately human-derived and, with sufficient expertise, decipherable. Their evaluation functions were built on handcrafted concepts familiar to any chess player: material advantage, pawn structure, king safety, and piece mobility.[3] One could dissect an engine's analysis and understand *why* it preferred a certain move based on these intelligible parameters. The modern NNUE engine operates on an entirely different principle. It learns not from human instruction but from self-play, processing millions of games to identify subtle, high-dimensional patterns that elude human cognition.[6] The engine's "thought process" occurs within a labyrinth of interconnected nodes and weights across multiple "hidden layers" in the neural network. While the input (a chess position) and the output (a move and an evaluation score) are clear, the intricate cascade of non-linear interactions that connects them is profoundly opaque.[6] This is the essence of the so-called "black box problem".[3] Even the developers of these world-beating systems can only describe in general

terms what the models do with the data they ingest; the specifics of their decision-making remain a mystery.[8]

The "Inhuman" Move and the Pedagogical Crisis

The practical consequence of this opacity manifests in what has become a common experience for even the strongest human players: the "inhuman" move. Grandmasters analyzing a position with a top NNUE engine are frequently confronted with suggestions that defy centuries of established chess strategy. The engine might propose a quiet, seemingly passive move in a sharp position, or a long-term piece sacrifice with no immediate, visible compensation.[9] These moves are often met with initial disbelief, only to be vindicated 15 or 20 moves later as their profound strategic depth becomes crushingly apparent.[11] While awe-inspiring, this phenomenon creates a significant barrier to learning. The engine provides the correct answer but offers no explanation, no strategic narrative that a human can internalize.

This has precipitated a pedagogical crisis in the chess world. The most powerful analytical tools in the history of the game are paradoxically becoming the most difficult to learn from. The traditional learning loop for a chess player involves playing a game, analyzing it to identify mistakes, understanding the conceptual reason for those mistakes, and integrating that new understanding to improve future play. The black box engine disrupts this crucial cycle at the "understanding" stage. A player is told their move was an error and is shown the superior alternative, but the underlying strategic lesson—the *why*—is missing. The process of learning risks being reduced from the acquisition of deep, conceptual knowledge to the rote memorization of engine-approved lines. The engine has transitioned from a powerful-but-understandable teacher into a cryptic oracle, delivering infallible pronouncements without ever revealing the source of its wisdom.[5]

The Alienation of the Expert

This inscrutability fosters a sense of alienation, particularly among the very experts who have dedicated their lives to mastering the game's complexities. A grandmaster's intuition is built upon a vast, shared library of patterns and strategic principles refined over generations. NNUE engines frequently operate outside this framework, identifying and exploiting patterns that are invisible to the human eye. When an engine's recommendation seems to violate a fundamental positional principle yet is proven to be superior, it challenges the expert's core

understanding and highlights the cognitive limits of the human mind in the face of the machine's alien logic. The analytical tool, intended to augment human skill, instead creates a chasm between human intuition and the "correct" chess revealed by the silicon oracle. This has created a powerful, unmet need within the elite chess community: not for an engine that is marginally stronger, but for one that is profoundly *wiser*—an engine that can begin to share the secrets of its knowledge, not just the conclusions of its calculations. It is this challenge that ShashChess has chosen to address.

Section 2: A Different Path: The Philosophy of ShashChess

In a landscape dominated by the relentless pursuit of marginal Elo gains, ShashChess represents a deliberate and significant philosophical divergence. While it proudly stands on the shoulders of giants, its developmental trajectory is guided by a unique mandate that prioritizes not just strength, but also relevance and insight for the human player.

Built on the Best, Guided by a Theory

At its core, ShashChess is a derivative of the world-champion Stockfish engine family.[13] It inherits the unparalleled speed and efficiency of Stockfish's alpha-beta search algorithm and its sophisticated implementation, ensuring a foundation of world-class playing strength from the outset. This lineage provides immediate credibility and guarantees that its analytical output is grounded in state-of-the-art chess programming. However, ShashChess is not a mere clone. Its primary and defining purpose is to serve as a vehicle for implementing the chess theories of Alexander Shashin, as detailed in his book, *Best Play: A New Method for Discovering the Strongest Move*.[15] This is not a superficial modification but a foundational principle that informs the engine's architecture and behavior. The project's goal from its inception has been to translate Shashin's human-centric strategic concepts into the language of computer code, aiming to create an engine that "thinks" in a way that is more resonant with a human master's perspective.

This philosophy is most clearly embodied in the engine's long-standing "personality" feature. Based on its evaluation of a position, ShashChess classifies the strategic landscape according to the styles of three legendary World Champions: the attacking genius Mikhail Tal, the strategic and endgame virtuoso José Raúl Capablanca, and the defensive master Tigran

Petrosian.[17] The engine then dynamically adjusts its search parameters and evaluation heuristics based on whether the position calls for attack (Tal), strategic maneuvering (Capablanca), or prophylaxis and defense (Petrosian).[19] This feature is a clear testament to the project's commitment to bridging the gap between a raw numerical evaluation and a meaningful, understandable strategic context. It demonstrates a pre-existing focus on making the engine's logic more transparent and useful to the practical player.

A Fork in Developmental Philosophy

The approach taken by ShashChess marks a crucial fork in the road of modern chess engine development. The mainline Stockfish project has achieved its unprecedented strength through a ruthlessly efficient and empirical development model centered on the Fishtest framework.[7] This distributed testing platform plays hundreds of thousands of games to statistically validate minuscule code changes. A proposed patch is accepted if, and only if, it demonstrates a measurable Elo gain.[1] This process is brilliantly effective but entirely agnostic to human comprehension. A change that makes the code more opaque or "incomprehensible" will be adopted without hesitation if it wins more games.[21] The ultimate metric is performance, not explainability.

ShashChess, while still valuing strength, operates under a different set of priorities. Its development is guided by a central, human-authored theory. This creates a distinct value proposition for the end-user. While Stockfish offers maximal *correctness*, ShashChess aims to provide a blend of correctness and conceptual *relevance*. This is reflected in the community's perception of the engine. Forum discussions among strong players and engine enthusiasts often highlight that ShashChess, while perhaps not the absolute strongest in short time control matches, is a powerful and interesting analysis tool that frequently suggests different, thought-provoking candidate moves and strategic plans compared to its parent engine.[18] Players use it not just to find the "best" move, but to generate ideas and gain a different perspective on a position.[18]

From this perspective, the introduction of a sophisticated eval trace is not an abrupt change in direction but the ultimate and logical expression of the ShashChess philosophy. It represents the next evolutionary step in the engine's long-standing mission: to peel back the layers of complexity and make the machine's thinking more transparent, more instructive, and more valuable as a collaborative partner in the human quest for chess understanding.

Section 3: The Trace: A Window into the Neural Mind

For decades, the primary interface between a chess engine and a human analyst has been a single, deceptively simple number: the evaluation score. A value like +0.51 has become the universal language of computer chess analysis, indicating a slight advantage for White, equivalent to roughly half a pawn.[23] This numerical score, while an invaluable guide, is a radical compression of information. It represents the final output of a deeply complex calculation, but it reveals nothing about the specific positional factors, piece relationships, and potential plans that contributed to it. It is the answer, delivered without the work. The new, sophisticated trace feature in ShashChess, accessible via the eval command, is designed to be the first major step in unpacking that number and revealing the rich analytical detail hidden within.

Beyond the Evaluation Score: The "Activity Order"

The core of this groundbreaking feature is the "Activity Order," a component of the trace output that provides a ranked list of every piece on the board for both White and Black. This list is ordered not by the pieces' material value, but by their "activity" as perceived by the engine's neural network. This functions as a "mini thinking system," offering an unprecedented, real-time glimpse into the engine's strategic and dynamic priorities.

It is crucial to understand that "activity" in this context is a far more sophisticated concept than simple mobility—the raw count of squares a piece can legally move to.[23] The NNUE's assessment is holistic and context-aware. It evaluates a piece's dynamic potential, its role in coordinating with other pieces, its control of critical squares, its participation in attacking or defensive structures, and its potential to become stronger or weaker in the ensuing variations. A rook on an open file might rank highly, as expected. But a bishop on its starting square could also rank highly if the NNUE determines it is a critical defender of the king or that it will become a monster piece after an impending pawn break. Conversely, a queen developed to the center of the board might receive a low activity rank if the network perceives it to be exposed, lacking targets, or a future liability. The Activity Order, therefore, reveals the engine's judgment on the *usefulness* and *potential* of each piece within the specific context of the position.

The Decryption Key in Action: A Comparative Trace Analysis

The true power of this feature is unlocked when used as a decryption key—a tool to translate

the alien language of the neural network into the familiar grammar of human chess strategy. To demonstrate this, the feature is best utilized in tandem with ShashChess's sibling engine, Alexander. Alexander is derived from a classical version of Stockfish and uses a traditional, handcrafted evaluation function built on understandable chess concepts.[4] By analyzing the same position with both engines, we can create a powerful comparative framework.

Consider the following complex middlegame position, analyzed using this dual-engine approach:

The Position: A Tense Middlegame	**ShashChess NNUE Trace Output**	**Alexander Classical Trace Output**
Diagram: *(Note: A representative diagram would be inserted here)* **FEN:** r1b2rk1/pp3ppp/1q2p3/2npP3/5Q2/2N2N2/PPP2PPP/R4RK1 w - - 2 15 **Description:** White has established a strong knight on e5 and exerts pressure on the kingside. Black's pieces are solidly placed, with the knight on c5 controlling key squares and the queen on b6 eyeing White's queenside weaknesses. The position is dynamically balanced and rich in strategic possibilities.	**Evaluation:** +0.48 **Shashin Zone:** Tal-Capablanca **White Activity Order:** 1. Ne5 (Central Control, Attacking Potential) 2. Qf4 (King Pressure, Active Piece) 3. Rf1 (Preparing to join attack via f3) 4. Nf3 (Supporting e5, Defending King) 5. Nc3 (Controlling Center) 6. Ra1 (Potential for a4-a5 or swinging to kingside) 7. g1 (King) 8. g2 (Pawn) ... (other pawns) **Black Activity Order:** 1. Nc5 (Key Positional Piece, Queenside Pressure) 2. Qb6 (Active, Threatening Pawns) 3. f8 (Rook, Defending Kingside) 4. g8 (King) 5. ...	**Evaluation:** +0.41 **White Evaluation Terms (MG/EG):** - **Space:** +18 / +10 - **King Attack:** +25 / +5 - **Mobility:** +12 / +15 - **Piece Coordination:** +8 / +6 **Black Evaluation Terms (MG/EG):** - **Pawn Structure:** +5 / +12 - **Passed Pawns:** 0 / +4 **Summary:** White's advantage is primarily based on superior space and mounting pressure against the Black king. Black's compensation lies in a slightly better pawn structure.

This comparative table acts as a Rosetta Stone. A human analyst, looking at the ShashChess output alone, sees that the knight on e5 is ranked as the most important piece. This is a powerful clue, but the reason remains implicit. By cross-referencing with the Alexander trace, the reason becomes explicit. The largest single contributor to White's advantage in Alexander's classical evaluation is the "King Attack" term, with a score of +25 centipawns. The connection is immediate and illuminating: the NNUE prioritizes the Ne5 because it correctly identifies it as the linchpin of the entire attacking structure.

Similarly, the NNUE ranks Black's knight on c5 as its most active piece. The Alexander trace doesn't have a single term for "queenside pressure," but a human analyst can synthesize this from the data. The Nc5 is what holds Black's position together, blunts White's initiative on the queenside, and poses long-term threats. The NNUE has distilled this complex set of strategic functions into a single, high-ranking placement in its Activity Order. This process transforms the black box into a glass box. It allows the analyst to peer inside the neural network's decision-making process, using the familiar concepts of classical evaluation as a guide to understanding the NNUE's deeper, more nuanced positional judgments.

Section 4: The Rosetta Stone: A Practical Framework for Explainable AI in Chess

The capacity to decipher the internal priorities of a neural network engine elevates the ShashChess and Alexander pairing beyond a mere set of analytical tools. It establishes a practical and powerful framework for what is known in the broader technology landscape as Explainable AI (XAI). XAI is a rapidly emerging field dedicated to addressing the black box problem across numerous domains, with the goal of building artificial intelligence systems that are transparent, interpretable, and accountable.[25] In high-stakes applications such as medicine, finance, and autonomous vehicles, the ability to understand *why* an AI made a particular decision is not a luxury, but a necessity for trust, validation, and responsible deployment.[27] An AI that diagnoses a disease or denies a loan application without providing a clear rationale is of limited practical use, as its conclusions cannot be verified or challenged by human experts.[8]

The challenges faced in chess are a direct parallel. An engine that recommends a move without offering any insight into its strategic basis fails to transfer its knowledge effectively, limiting its utility as a tool for human improvement. The ShashChess/Alexander system represents a novel, domain-specific methodology for achieving explainability in the complex world of chess.

A Unique, Hybrid XAI Methodology

Most XAI techniques fall into two broad categories. *Post-hoc* methods attempt to explain a pre-existing black box model from the outside, often by approximating its complex decision-making with a simpler, more interpretable model (like LIME or SHAP).[26] *Ante-hoc*

methods, by contrast, involve building models that are inherently interpretable from the ground up, though this often requires a trade-off that can compromise predictive accuracy.[26] The ShashChess/Alexander framework pioneers a third path—a hybrid, comparative approach that leverages the strengths of both neural and symbolic AI paradigms.

This system functions by creating a dialogue between two distinct forms of machine intelligence:

1. **ShashChess (The Neural Intuition):** The primary engine, equipped with its full-strength NNUE, provides a *local, model-specific* explanation via the eval trace.[26] The Activity Order answers the crucial question: "**What** does the AI consider most important in this specific position, right now?" It reveals the focal points of the network's attention without immediately explaining the underlying logic in human terms.
2. **Alexander (The Classical Logic):** The secondary engine, using its transparent, hand-crafted evaluation function, provides a *global, interpretable model* of chess strategy.[5] Its trace, composed of familiar terms like "Space," "King Safety," and "Mobility," answers the complementary question: "**Why** might these focal points be important, according to established chess principles?"

By cross-referencing the "what" from ShashChess with the "why" from Alexander, the human analyst can construct a coherent and insightful narrative. This process effectively builds a bridge from the NNUE's opaque, high-dimensional pattern recognition to a human-understandable strategic reason. This methodology is powerful because it sidesteps the classic XAI dilemma of sacrificing accuracy for interpretability.[27] It does not require simplifying or altering the state-of-the-art NNUE model in any way. Instead, it uses a second, fully transparent model as an interpretive lens, a decoder key to unlock the insights of the first. This represents a significant and practical contribution to the field of applied XAI, creating a system where the whole is far greater than the sum of its parts. It is a "human-in-the-loop" explanation system, empowering the analyst to use the two tools in concert to forge a deeper understanding, transforming the engine from an inscrutable oracle into a collaborative and transparent partner.

Section 5: From Theory to Practice: Applications for the Elite Player and Analyst

The theoretical power of this new XAI framework translates into a set of concrete, high-value applications that directly address the most pressing challenges faced by elite chess players, coaches, and analysts in the modern era. The eval trace feature is not an academic curiosity; it is a practical weapon for competitive preparation and a revolutionary tool for chess

education.

Use Case 1: Deciphering Opening Novelties

A grandmaster is preparing for a critical tournament game and uses an engine to check their opening preparation. In a key line, the engine suggests a surprising novelty—a quiet developing move where a more aggressive, tactical continuation was always considered standard. Previously, the GM would be left to wonder about the move's deep purpose, perhaps trusting the engine's verdict without fully grasping its intent.[30] With the new trace, the process is transformed. By analyzing the position after the novelty, the GM sees that the seemingly passive move dramatically elevates the "Activity Order" ranking of a key piece, for example, a bishop that now has an unobstructed diagonal that will become critical 10 moves later. The Alexander trace might confirm this by showing a significant long-term increase in the "Mobility" or "Piece Coordination" score. The GM now understands the deep strategic concept behind the novelty and can wield it with confidence, knowing not just *what* to play, but *why* it is strong.

Use Case 2: Navigating "Quiet" Middlegames

One of the most difficult tasks for a human player is to convert a small, persistent positional advantage in a position with no obvious tactics. The engine may display a steady +0.40 evaluation, but the source of this advantage can be completely invisible to the human eye, leading to aimless play and a squandered edge.[5] The ShashChess trace provides an immediate focal point. It might reveal that the entire advantage is predicated on the activity of a single piece—for instance, a rook whose activity rank is 1 while all other pieces are ranked much lower. This tells the player that the correct plan is to play around this specific piece, to improve its scope, and to use it as the fulcrum for all future operations. The trace cuts through the fog of a complex position to provide a clear, actionable strategic directive.

Use Case 3: A New Paradigm in Game Review

Post-game analysis is the cornerstone of chess improvement. Currently, this process is dominated by metrics like "centipawn loss" or labels like "Inaccuracy" and "Blunder".[31] These

are useful for identifying errors but are poor at explaining the nature of the mistake. A player might see they lost 0.7 pawns of evaluation but struggle to understand the positional reason. The trace feature enables a far deeper and more instructive form of review. Instead of a simple numerical penalty, the analysis can provide a conceptual explanation. For example: "Your move, ...Re8, was an inaccuracy because it left your light-squared bishop as your least active piece, with a rank of 8. The engine's preferred move, ...b5, would have immediately activated this bishop by opening the c8-h3 diagonal, raising its activity rank to 3." This provides a direct, unforgettable lesson in positional play, teaching the player to think not just in terms of good or bad moves, but in terms of piece activity and potential.

Use Case 4: Advanced Opponent Modeling

Top-level preparation involves not just analyzing openings, but analyzing the opponent. The trace feature allows for an unprecedented level of depth in this process. By running a collection of an opponent's games through the ShashChess analyzer, a player and their coach can identify recurring patterns in their strategic thinking. Does the opponent consistently activate their knights well but neglect their rooks? Do they excel in positions where their queen has a high activity rank but struggle when their defensive pieces are most critical? This moves beyond knowing an opponent's opening repertoire to understanding their cognitive biases and strategic blind spots. It allows for preparation that targets the opponent's specific psychological and strategic weaknesses, a level of insight previously available only through years of over-the-board experience.

Ultimately, these applications signal a fundamental shift in the goal of human-computer interaction in chess. The old model was one of *validation*: "I have a strategic idea; let me use the engine to check if it's tactically sound." The opacity of black box engines made it difficult to move beyond this dynamic. The new model enabled by the ShashChess trace is one of *discovery*: "The engine has a deep strategic idea; let me use the trace to understand it." This transforms analysis from a chore of error-checking into a collaborative exploration, empowering the human player to learn from the machine's alien genius and discover new strategic frontiers.

Section 6: The Next Move: The Future of Human-Computer Collaboration in Chess

The evolution of chess engines is approaching a critical juncture. For years, the primary

metric of progress has been the Elo rating—a relentless arms race to create the strongest possible playing entity. While this pursuit has yielded machines of breathtaking power, it is also reaching a point of diminishing returns for the vast majority of the chess-playing world. The practical difference between an engine rated 3600 and one rated 3620 is negligible for even the strongest grandmasters. The future of engine development, particularly for the crucial purpose of analysis and learning, lies not in adding a few more Elo points, but in deepening the partnership between human intuition and machine calculation. It is a future defined by collaboration, not just computation.

ShashChess, with the introduction of its sophisticated eval trace, is at the vanguard of this new movement. This feature is more than a technical upgrade; it is a statement of intent. It asserts that the next great leap forward in computer chess will be in the realm of explainability. It is a commitment to transforming the engine from an unfeeling calculator into a true intellectual partner—one that can not only provide answers but can also illuminate the path to understanding them.

An Invitation to a New Field of Discovery

This document and the release of the new ShashChess are intended as the beginning of a new and vital conversation. The challenge of fully understanding the strategic mind of a neural network is monumental, akin to the work of neuroscientists mapping the complexities of the human brain. It is a task too large for any single developer or research group to tackle alone. Therefore, we extend a formal invitation to the global chess community to participate in this new chapter of discovery.

We invite grandmasters and international masters to use this tool to dissect their games and preparation, to uncover the hidden logic behind baffling engine novelties, and to share their findings with the world. We invite the dedicated and brilliant programmers and theorists who populate forums like TalkChess [18], the Computer Chess Club [33], and Lichess [22] to test, critique, and build upon this framework. By releasing this tool, we are effectively launching a massive, distributed research project. Every time an analyst uses the trace to understand a complex position and shares their findings, they contribute a small piece to the collective puzzle. Over time, this crowdsourced effort will build a qualitative, community-driven understanding of the NNUE's priorities. A new, collaborative field of study will emerge: "Applied NNUE Chess Neuro-strategy."

A Call to Action: Join the Exploration

The tools for this exploration are open, free, and available to all. We encourage every serious chess player, analyst, and developer to take the next step:

- **Download the Engines:** The latest official releases of both ShashChess and its classical counterpart, Alexander, are available for Windows, Linux, and Android systems on the official project GitHub.
  - **ShashChess:**(https://github.com/amchess/ShashChess/releases) [34]
  - **Alexander:** https://github.com/amchess/Alexander [24]
- **Join the Discussion:** Share your analyses, questions, and discoveries in the central hub for computer chess discussion. The official ShashChess development thread on the TalkChess forum is the primary venue for engaging with the author and a community of experts.
  - **TalkChess Forum:** https://talkchess.com/ [18]
- **Explore the Code:** ShashChess is an open-source project distributed under the GNU General Public License.[14] We invite developers to examine the implementation of this new feature, particularly within the trace.cpp and trace.h source files. Learn from it, experiment with it, and perhaps contribute to the future of explainable AI in chess.
  - **ShashChess Source Code:**(https://github.com/amchess/ShashChess) [14]

The black box has been the defining challenge of the modern chess era. With this release, we are providing the first key to unlocking it. The journey of decryption has just begun.

Bibliografia

1. Stockfish (chess) - Wikipedia, accesso eseguito il giorno ottobre 18, 2025, https://en.wikipedia.org/wiki/Stockfish_(chess)
2. Strongest Chess Engines (2025) - Reddit, accesso eseguito il giorno ottobre 18, 2025, https://www.reddit.com/r/chess/comments/1nalduq/strongest_chess_engines_2025/
3. Using Regression To Understand the AlphaZero Chess Engine | Towards Data Science, accesso eseguito il giorno ottobre 18, 2025, https://towardsdatascience.com/using-regression-to-understand-the-alphazero-chess-engine-aba67148978b/
4. Stockfish - Chessprogramming wiki, accesso eseguito il giorno ottobre 18, 2025, https://www.chessprogramming.org/Stockfish
5. Understanding Engines 1 - Chess Journal, accesso eseguito il giorno ottobre 18, 2025, https://www.chess-journal.com/classicalEngines.html
6. Solving the Black Box Problem: A General-Purpose Recipe for Explainable Artificial Intelligence Carlos Zednik Otto-von-Guericke- - arXiv, accesso eseguito il giorno ottobre 18, 2025, https://arxiv.org/vc/arxiv/papers/1903/1903.04361v1.pdf
7. I'm a Stockfish/Leela Chess Zero Developer. Ask me anything! - Reddit, accesso eseguito il giorno ottobre 18, 2025,

https://www.reddit.com/r/chess/comments/1jtctnq/im_a_stockfishleela_chess_zero_developer_ask_me/
8. Understanding Black Box AI: Challenges and Solutions - EWSolutions, accesso eseguito il giorno ottobre 18, 2025, https://www.ewsolutions.com/understanding-black-box-ai/
9. Lc0 vs. Stockfish - Chess Stack Exchange, accesso eseguito il giorno ottobre 18, 2025, https://chess.stackexchange.com/questions/32024/lc0-vs-stockfish
10. Stockfish defeats Leela C0 and prooves that AlphaZero was a myth - Chess Forums, accesso eseguito il giorno ottobre 18, 2025, https://www.chess.com/forum/view/general/leela-c0-is-crushing-stockfish-uncle-alphazero-is-proud-and-fanboys-are-crying-under-de-bed
11. Stockfish 17 DESTROYS Leela C Zero in 29 Moves!? - YouTube, accesso eseguito il giorno ottobre 18, 2025, https://www.youtube.com/watch?v=osRrO7_VRzM&pp=0gcJCfwAo7VqN5tD
12. Engines vs Grandmasters - Chess Forums, accesso eseguito il giorno ottobre 18, 2025, https://www.chess.com/forum/view/chess-openings/engines-vs-grandmasters
13. ShashChess Pro 1.0 64-bit in CCRL, accesso eseguito il giorno ottobre 18, 2025, https://computerchess.org.uk/ccrl/402.archive/cgi/engine_details.cgi?print=Details&each_game=0&eng=ShashChess%20Pro%201.0%2064-bit
14. amchess/ShashChess: A try to implement Alexander Shashin's theory on a Stockfish's derived chess engine - GitHub, accesso eseguito il giorno ottobre 18, 2025, https://github.com/amchess/ShashChess
15. Introducing ShashChess: A New Addition to the Chessify Engine Suite, accesso eseguito il giorno ottobre 18, 2025, https://chessify.me/news/shashchess-engine-on-chessify-cloud
16. Engine Request: Shashchess - Next Chess Move, accesso eseguito il giorno ottobre 18, 2025, https://forums.nextchessmove.com/t/engine-request-shashchess/1006
17. ShashChess - Chessprogramming wiki, accesso eseguito il giorno ottobre 18, 2025, https://www.chessprogramming.org/index.php?title=ShashChess&mobileaction=toggle_view_desktop
18. ShashChess - Page 27 - TalkChess.com, accesso eseguito il giorno ottobre 18, 2025, https://talkchess.com/viewtopic.php?t=78738&start=260
19. ShashChess Parameters - Next Chess Move, accesso eseguito il giorno ottobre 18, 2025, https://forums.nextchessmove.com/t/shashchess-parameters/1188
20. Creating chess engine, machine learning vs. traditional engine? - Chess Stack Exchange, accesso eseguito il giorno ottobre 18, 2025, https://chess.stackexchange.com/questions/26489/creating-chess-engine-machine-learning-vs-traditional-engine
21. What is the strongest chess engine in the world? — A Reflection - Page 2 - TalkChess.com, accesso eseguito il giorno ottobre 18, 2025, https://talkchess.com/viewtopic.php?t=84490&start=10
22. New Engine: ShashChess 38: Stockfish Derived Engine - Lichess, accesso

eseguito il giorno ottobre 18, 2025, https://lichess.org/forum/general-chess-discussion/new-engine-shashchess-38-stockfish-derived-engine
23. Behind the Numbers: Understanding Chess Engine Evaluations - Chessify, accesso eseguito il giorno ottobre 18, 2025, https://chessify.me/blog/chess-engine-evaluation
24. amchess/Alexander: A uci chess engine derived from ... - GitHub, accesso eseguito il giorno ottobre 18, 2025, https://github.com/amchess/Alexander
25. What is Explainable AI (XAI)? - IBM, accesso eseguito il giorno ottobre 18, 2025, https://www.ibm.com/think/topics/explainable-ai
26. Explainable artificial intelligence (XAI): from inherent explainability to large language models - arXiv, accesso eseguito il giorno ottobre 18, 2025, https://arxiv.org/html/2501.09967v1
27. What challenges do Explainable AI systems face in highly complex domains? - Milvus, accesso eseguito il giorno ottobre 18, 2025, https://milvus.io/ai-quick-reference/what-challenges-do-explainable-ai-systems-face-in-highly-complex-domains
28. Top Use Cases of Explainable AI: Real-World ... - SmythOS, accesso eseguito il giorno ottobre 18, 2025, https://smythos.com/developers/agent-development/explainable-ai-use-cases/
29. Explainable artificial intelligence - Wikipedia, accesso eseguito il giorno ottobre 18, 2025, https://en.wikipedia.org/wiki/Explainable_artificial_intelligence
30. Which Chess Engine is Most Preferred by Grandmasters for Analysis and Considered the Fastest and Best Currently?, accesso eseguito il giorno ottobre 18, 2025, https://chess.stackexchange.com/questions/43565/which-chess-engine-is-most-preferred-by-grandmasters-for-analysis-and-considered
31. Been tinkling with Chess Engine calculation, I'm curious how does Stockfish compute accuracy at the end of a Chess.com game? - Reddit, accesso eseguito il giorno ottobre 18, 2025, https://www.reddit.com/r/chess/comments/1g3lezj/been_tinkling_with_chess_engine_calculation_im/
32. TalkChess.com - Index page, accesso eseguito il giorno ottobre 18, 2025, https://talkchess.com/
33. Computer Chess Forums - Chessprogramming wiki, accesso eseguito il giorno ottobre 18, 2025, https://www.chessprogramming.org/Computer_Chess_Forums
34. Releases · amchess/ShashChess - GitHub, accesso eseguito il giorno ottobre 18, 2025, https://github.com/amchess/ShashChess/releases